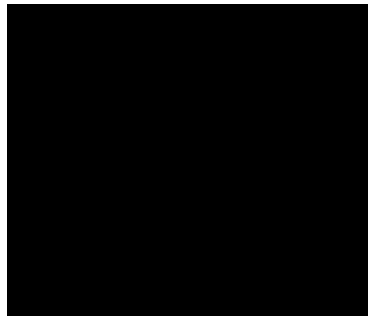


# Project Specifications and Risk Assessment

## Baldr Security System

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# 1 High Level Description

## 1.1 Motivation

In 2014 there were over 150,000 cases of breaking and entering across Canada [1]. Cottages, farms, work sites, and other remote properties are prime targets for these crimes for a multitude of reasons. These properties are often too isolated for the incident to be noticed by any bystanders, too remote for effective law enforcement response, and may be vacant for weeks or months at a time. Generally, such properties are not very secure and contain very valuable objects, including personal belongings, electronics, and possibly location based vehicles such as boats, ATVs, or farm equipment. Due to the isolation of such properties almost anything can be a reasonable target as an intruder will have plenty of time to remove valuable items.

Security systems are not able to prevent someone from breaking into a property, but they can detect when something has gone wrong and allow the owner to respond in a timely manner. This may not be enough to allow the police to arrive in time to make an arrest, but it may force the intruder to leave sooner and allow the police to arrive before any evidence is destroyed. Being informed of a door or window being opened while the owner is away, whether by an intruder, an animal, or a tree branch, also enables the owner to take action before further property damage is caused by weather, or animals entering the building. Even if nothing ever goes wrong, knowing that their property is secure and undamaged for their next visit can bring an owner peace of mind.

## 1.2 Project Objective

The objective of this project is to provide the owners of remote properties with a security system which will detect when a change occurs to a building on their property and notify them of the change. This will make owners aware of the state of their property and give them the opportunity to respond to events in a timely manner. Due to the remoteness of the target properties the system is designed to: run off of batteries so that it does not require access to the electrical grid, communicate with the property owner over the cellular network so it does not require an internet connection, and operate in a wide range of temperatures so that it can be used in unheated buildings throughout Canada.

## 1.3 Block Diagram

The Baldr Security System is a battery powered security system containing a network of sensors, cameras, and a central modem capable of cellular communication. When a sensor is triggered by a change in or around the building, the system activates the cameras and sends images to the owner via the cellular network to notify them of the situation on their property. Figure 1 below illustrates the four subsystems which will make up the proposed design solution, along with the individual components found within them and their connections to one another.

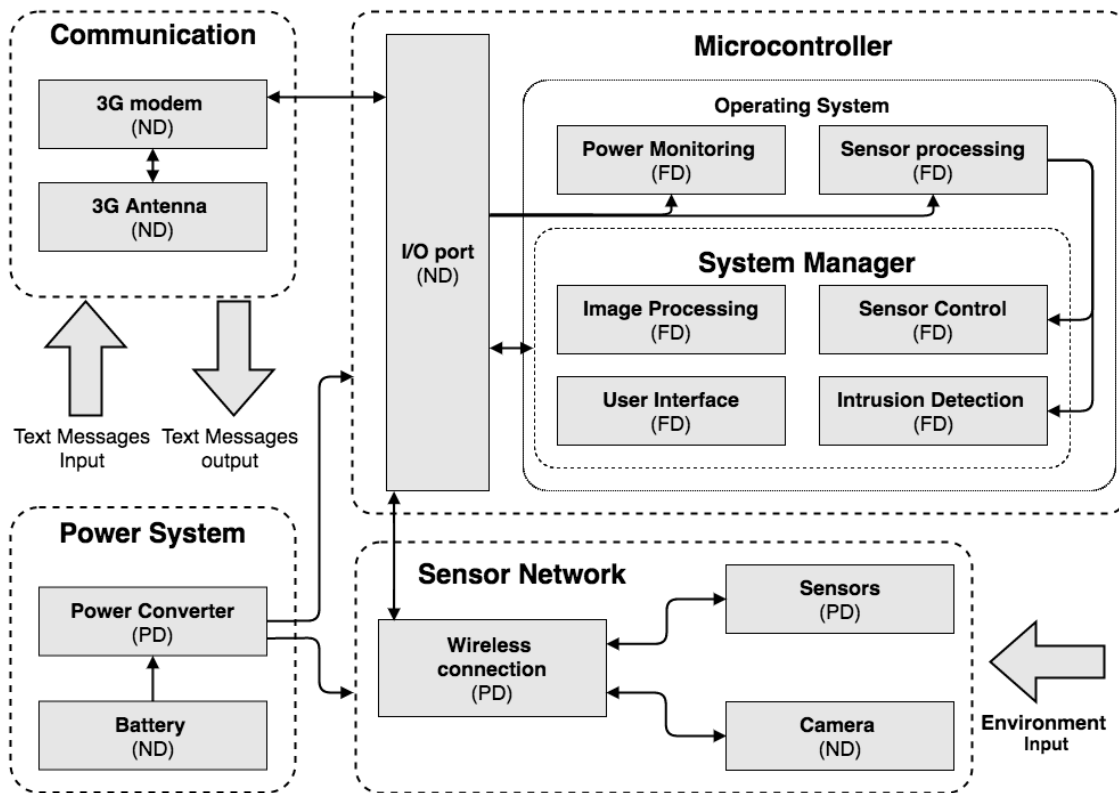


Figure 1: Block diagram of the low-power security system showing connections between subsystems. Blocks mark as (FD) will be fully custom designed, blocks marked as (PD) will be partially custom designed, and blocks marked as (ND) will not have any custom design.

### 1.3.1 Power Subsystem

The power system will supply power to the microcontroller subsystem, and will require a power converter in order to do so. The components for this subsystem will make use of existing technologies, however, the selection and assembly of components will require some design in order to meet the system's specifications. Table 1 below gives a detailed description of each component in this subsystem.

Table 1: Detailed description of the power subsystem's components

| Component       | To Be Designed | Description                                                                                                    |
|-----------------|----------------|----------------------------------------------------------------------------------------------------------------|
| Battery         | No             | Provides power for all subsystems using standard replaceable consumer batteries such as AA and coin cells.     |
| Power Converter | Partially      | DC to DC power regulator to supply the required power to the other subsystems in a wide range of temperatures. |

### 1.3.2 Communications Subsystem

The communications subsystem is an essential part of the overall design, but will make use of the existing cellular network towers in the area, along with 3G modem technologies currently obtainable on the market. Table 2 below gives a description of each component in this subsystem.

*Table 2: Detailed description of the communication subsystem's components*

| Component  | To Be Designed | Description                                                                            |
|------------|----------------|----------------------------------------------------------------------------------------|
| 3G Antenna | No             | Connects the 3G Modem to the cellular network                                          |
| 3G Modem   | No             | Sends images and text to the user and receives messages back over the cellular network |

### 1.3.3 Microcontroller Subsystem

The microcontroller will handle all sensor and camera information as well as any interaction with the user. Decision algorithms are applied to sensor data to determine whether any activity noted by the sensors should be considered an intrusion. In such a situation, the cameras are instructed to power up and take an image of the surrounding area. When the microcontroller receives the image it will compress it and send it to the user. In addition to processing information from the sensor subsystem, the microcontroller subsystem will also monitor the power subsystem for remaining battery life. Table 3 below gives a description of each component in this subsystem.

*Table 3: Detailed description of the microcontroller subsystem's components*

| Component           | To Be Designed | Description                                                                                                                                                            |
|---------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I/O Ports           | No             | A bus that passes commands and data between each of the subsystems                                                                                                     |
| Power Monitoring    | Yes            | Monitors power system to estimate battery levels for system management purposes and to report battery life to the user                                                 |
| Sensor Processing   | Yes            | Reads raw sensor inputs and processes them to be easily used by the other software subsystems                                                                          |
| Image Processing    | Partially      | Captures images from the camera and compresses the data for transmission to the user                                                                                   |
| Sensor Control      | Yes            | Controls the polling frequency of each sensor and monitors the status of the sensor network                                                                            |
| Intrusion Detection | Yes            | Use sensor input to decide when notable activity has occurred                                                                                                          |
| User Interface      | Yes            | Send user a picture when there has been a problem detected and allow the user to give the system commands such as to take a picture on demand or arm/disarm the system |

### 1.3.4 Sensor Network Subsystem

The sensor network subsystem will consist of sensors for detecting changes in the environment of the building that has been isolated, and cameras for capturing the situation in the event that the system has determined it necessary for the user to be informed. Table 4 below gives a description of each component in this subsystem.

*Table 4: Detailed description of the sensor network subsystem's components*

| Component           | To Be Designed | Description                                                                              |
|---------------------|----------------|------------------------------------------------------------------------------------------|
| Sensors             | No             | Environment sensors that trigger the system and cause it to take pictures.               |
| Camera              | No             | Captures an image to allow the user to assess the situation when the system is triggered |
| Wireless Connection | Yes            | Wireless digital interface between remote sensor modules and main system                 |

## 2 Project Specifications

### 2.1 Functional Specifications

Table 5 gives a list of the functional specifications required for the system.

*Table 5: Details of the essential functional specifications for the system*

| Specification                           | Location                           | Description                                                                                                                    |
|-----------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Battery life while passively monitoring | Power subsystem                    | System should be able to operate for 12 months while waiting for a sensor to be triggered                                      |
| Battery life while keeping user updated | Power subsystem                    | System should be able to provide a current image to the user every 15 minutes for at least 1 day                               |
| Cellular throughput                     | Communication subsystem            | Peak upload throughput should be greater than 200 kbps to allow uploading a 100 kB picture within 5 seconds in good conditions |
| Spectrum regulations                    | Communication, Wireless connection | Emissions must not exceed regulated power in the most important international markets: Canada, USA, China, EU                  |
| Intrusion detection reaction time       | Sensor subsystem                   | The main CPU should be alerted within 30 seconds if a door or window that has an intrusion sensor is opened                    |

|                               |                          |                                                                                                                        |
|-------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------|
| Motion detection              | Sensor subsystem         | Motion sensor should be able to trigger the camera if a person is within 5 m of the camera in the camera field of view |
| Temperature range             | Overall system           | System should remain operational from -30°C to 40°C for at least 1 month                                               |
| Module communication distance | Sensors                  | Sensors using the wireless sensor network should be able to communicate at distances of up to 25 m                     |
| Camera reaction time          | Camera, Image processing | Camera must capture the scene within 5 seconds of being triggered by a sensor                                          |
| User notification time        | Communication system     | User should be notified of an important event within 1 minute of the sensor being triggered                            |
| Peak power                    | Power subsystem          | Power supply must be able to provide the maximum amount of power required by the other subsystems                      |
| Frequency of user updates     | Communication system     | System must be able to update the user with at least 1 new photo every 1 minute if the user requests it                |
| Setup connection time         | Sensor network           | Within 1 minute of adding a new sensor node to the system it should indicate it has connected successfully             |

## 2.2 Non-functional Specifications

Table 6 gives a list of the non-functional specifications required for the system.

*Table 6: Details of the essential non-functional specifications for the system*

| Specification       | Location                          | Description                                                                                          |
|---------------------|-----------------------------------|------------------------------------------------------------------------------------------------------|
| Wireless security   | Communication, Wireless subsystem | Wireless communication needs to be encrypted to prevent malicious users attacking the wireless links |
| Reliability         | Overall system                    | Mean Time Between Failure (MTBF) must be greater than 10 years (excluding battery replacements)      |
| Replaceable battery | Power subsystem                   | Batteries connections are designed to allow for field replacement                                    |

## 2.3 Non-essential Functional Specifications

Table 7 gives a list of functional specifications that would improve the system but are not required for it to meet the project objective.

*Table 7: Details of the non-essential functional specifications for the system*

| Specification    | Location                 | Description                                                                                                                                       |
|------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Deterrent buzzer | Sensor subsystem         | Loud enough the animal or person who caused the issue can hear it if they are in the same, nearly silent, 5 m x 5 m room                          |
| Deterrent light  | Sensor network subsystem | Sensor nodes should be able to visibly blink an LED for a period of 10 minutes after being triggered                                              |
| Web interface    | User interface           | Website interface to allow the user to interact with the system including arming/disarming, setup, checking power levels, and testing the cameras |

## 2.4 Non-essential Non-functional Specifications

Table 8 gives a list of non-functional specifications that would improve the system but are not required for it to meet the project objective.

*Table 8: Details of the non-essential non-functional specifications for the system*

| Specification              | Location       | Description                                                            |
|----------------------------|----------------|------------------------------------------------------------------------|
| Weight per unit            | Overall System | Less than 10 Kg for each unit including battery                        |
| System physical dimensions | Overall system | Entire physical system should fit inside a 50 cm x 50 cm x 50 cm space |
| Cost                       | Overall system | Total parts cost for the system should be less than \$1000             |

## 3 Risk Assessment

Table 9 gives a list of the greatest risks involved in the design and creation of the system. Each risk is given an impact and likelihood score of either low, medium, or high based on how they will affect the project and how likely it is to occur. Each risk also has a possible mitigation strategy outlined for it.



Table 9: Description of project risks including their impact and likelihood, as well as mitigation plans

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                               | Impact | Likelihood |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------|
| Situation   | <b>Complexity too Significant for Time Frame</b>                                                                                                                                                                                                                                                                                                                                                                                              | High   | Medium     |
| Description | This is a significant risk that is tied to both skill and time management of the project. We are dealing with multiple different modules that require some knowledge or skills that a fourth year ECE student may not already have.                                                                                                                                                                                                           |        |            |
| Mitigation  | We are planning on prioritizing the main components of our system that are required to make the product functional, then move onto other aspects that make the design more marketable, like the aspect of sending photographic updates to the user in the case of intrusion. To ensure that we meet our deadlines, we will also ensure that each member of the group puts in at least 8 hours of research and work per week into the project. |        |            |
| Situation   | <b>Power Consumption Goals are too Difficult to Achieve</b>                                                                                                                                                                                                                                                                                                                                                                                   | High   | High       |
| Description | The power consumption goals for our project are a key factor that differentiates our security system from other security systems. Having only a single electrical engineer in the group limits our knowledge on power systems, which may lead to problems in the future.                                                                                                                                                                      |        |            |
| Mitigation  | We have been researching different design options and improvements for the power system that we can explore throughout the rest of the design phase. In addition, we are going to make an effort to refer to our consultant frequently, who is highly experienced with power systems and technology, giving us a push in the right direction for our system.                                                                                  |        |            |
| Situation   | <b>Purchasing Hardware</b>                                                                                                                                                                                                                                                                                                                                                                                                                    | Low    | High       |
| Description | Through the design of this project, we are going to need to purchase many electrical products and modules. We will need sensors, modems, low power cameras, external batteries and various electrical components such as microcontrollers and capacitors for the circuit design portion. Shipping items from multiple international sources has its risks, such as damages, delays and possible losses.                                       |        |            |
| Mitigation  | By isolating our purchasing to a reliable source such as DigiKey we will be able to lower these risks.                                                                                                                                                                                                                                                                                                                                        |        |            |

# References

- [1] Stats Canada, "Stats Canada," [Online]. Available:  
[http://www5.statcan.gc.ca/cansim/a26?lang=eng&retrLang=eng&id=2520051&pattern=&csid=.](http://www5.statcan.gc.ca/cansim/a26?lang=eng&retrLang=eng&id=2520051&pattern=&csid=)  
[Accessed 29 May 2016].